

MEMOPAD: Multimodal Emotion Monitoring in Clinical Pathways for Anxiety Disorders

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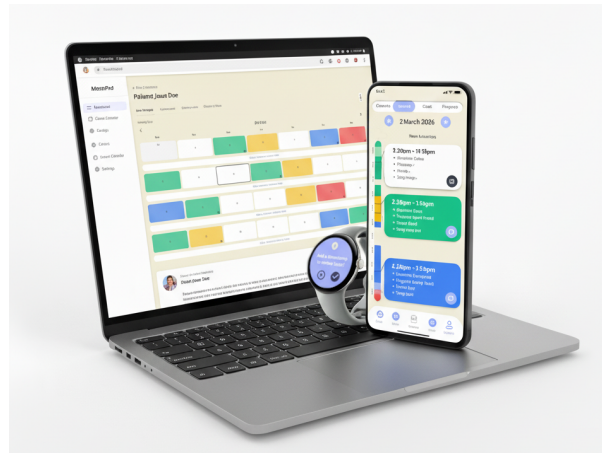


Figure 1: Overview of the MEMoPAD system components and user interface.

Abstract

MEMoPAD (Multimodal Emotion Monitoring in Clinical Pathways for Anxiety Disorders) is a wearable-based emotion monitoring system co-designed with people with lived experience of anxiety, unpaid carers, and mental health clinicians across 6 iterative phases (4 completed with 19-25 participants each, plus a preparation phase). The system combines continuous physiological sensing (electrodermal activity, heart rate, skin temperature) via a commercial smartwatch with a companion mobile app and clinician web dashboard. A key co-design insight drove the interaction design: participants rejected pathology-focused framing, leading to a four-color “emotion

zone” vocabulary that tracks the full emotional spectrum rather than monitoring anxiety alone. This demo enables visitors to experience the complete data ecosystem: features on the smartwatch to break anxiety loops, emotion data visualizations, and exploring the clinician-facing dashboard showing longitudinal patterns.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; • **Applied computing** → Health informatics.

Keywords

Anxiety Disorders, Affective Computing, Wearables, Co-design, Digital Mental Health, Emotion Recognition, Clinical Integration, Mental Health

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1 Introduction and background

Digital mental health interventions (DMHIs) face a well-documented paradox: while efficacy is demonstrated in controlled trials, real-world effectiveness remains constrained by high attrition rates, a problem characterized as the “Achilles heel” of digital therapeutics [13]. A recent meta-analysis of 176 randomized controlled trials found that mental health apps show promise for depression and anxiety symptoms, yet engagement typically collapses within weeks of initial use [7].

Anxiety disorders, including generalized anxiety disorder (GAD), panic disorder, social anxiety disorder, obsessive-compulsive disorder (OCD), and post-traumatic stress disorder (PTSD), affect approximately one in ten people in the UK [11]. These conditions present population-specific engagement barriers: the symptoms these tools aim to address—including avoidance behaviours, cognitive overload, and fear of judgment—may actively work against sustained use. Standard self-reported approaches rely paradoxically on patients’ ability to assess their own emotional awareness, a capability often impaired in anxiety disorders [2].

While wearable devices capable of detecting physiological markers of emotional states have matured considerably [9], and affective computing holds potential to enhance clinical pathways for anxiety disorders [10], there is still a lack of systems implemented in clinical pathways.

This gap motivated MEMoPAD (Multimodal Emotion Monitoring in Clinical Pathways for Anxiety Disorders): a wearable-based emotion monitoring system developed through a 6+1 phase co-design process with 19–25 participants per phase, including people with lived experience of anxiety, unpaid carers, and mental health clinicians.

1.1 Wearables for Mental Health

Consumer wearable devices (e.g. Google’s Pixel Watch2 [6]) now routinely capture physiological signals that can be relevant to detect emotional states. Continuous electrodermal activity (cEDA), heart rate activity, and skin temperature can serve as proxies for sympathetic nervous system arousal associated with anxiety [5].

However, translating these signals into clinically meaningful insights presents several open questions: which narrative would benefit the users? How should data be visualised to balance scientific rigor and user understanding? How should users’ self-reflection be supported without disrupting the therapeutic process of journaling?

Moreover, in affective computing, multimodal approaches combining multiple sensor streams typically outperform unimodal methods, achieving accuracy improvements of 10–20% [10]. Thus, MEMoPAD has been designed to be multimodal (Figure 2).

1.2 Co-design in Digital Mental Health

Co-design has been proposed as one approach to address the dropout problem endemic to DMHIs by involving end-users in development [4, 12]. Thieme et al. [14] demonstrated that human-centered AI design for mental health requires close collaboration with clinicians

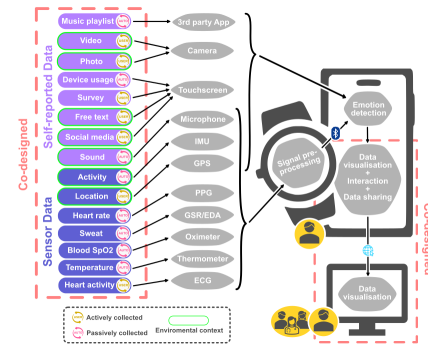


Figure 2: Overview of the multimodal MEMoPAD system design, and role of the different stakeholders: patients and supporters.

throughout development. However, few studies document how co-design with anxiety populations specifically shaped engagement-focused design decisions. This gap is significant because anxiety symptoms may actively undermine engagement with the very tools designed to help; a recursive challenge that generic user-centered design methods may not adequately address.

1.3 The Clinical Integration Gap

A literature review of 38 studies on affective computing applications for anxiety disorders found that while promising results have been demonstrated in research and digital environments, clinical integration remains rare [10]. The review identified a predominant focus on PTSD (55% of studies), with other anxiety subtypes underrepresented. Crucially, most systems were developed without substantial patient involvement in design decisions, and none reported systematic co-design with the triad of patients, carers, and clinicians across multiple iterative phases.

2 Our Co-design Process

2.1 Methodology Overview

MEMoPAD’s development follows a structured co-design framework adapted from Noorbergen et al. (2021) [12], spanning seven phases: an initial consultation phase (Phase 0) followed by six design phases. Phase V is currently underway. Phases I to IV each involved 19–25 participants across three stakeholder groups: people with lived experience of anxiety disorders, unpaid carers, and mental health clinicians from the NHS and private sector settings.

The approach was shaped by initial consultations (Phase 0, n=14), analysis of the literature, training materials from Health Innovation West of England, and guidance from interdisciplinary experts including a clinical psychologist. Each phase employs distinct Patient and Public Involvement and Engagement (PPIE) tools appropriate to its research questions. Data collection includes patients focus groups (2-hour sessions with up to six patient participants), separate one-to-one interviews with carers and clinicians (to mitigate power imbalances), and asynchronous feedback channels.

Analysis methods include thematic analysis (Phases I–II), affinity diagramming (Phases III–V), and comparative analysis across

stakeholder groups, followed by roundtable discussions with the interdisciplinary supervisory team (co-authors of this paper). A monthly newsletter, videos, and dedicated project website maintain ongoing engagement with participants between phases. Each phase begins with presentation of prior phase findings, ensuring continuity and enabling iterative refinement.

Phase V involved 16 participants in prototype testing under controlled lab conditions. Notably, 14 participants were new to MEMOPAD, yet they confirmed appreciation for features prioritized by earlier co-design cohorts. This suggests the engagement strategies may generalize beyond the original design participants (Table 1).

Phase VI (planned mid-2026) is designed to deploy an advance iteration of the prototype with consenting participants over 2-3 weeks 'in the wild' settings. Research questions focus on whether engagement persists in daily use, how real-time interventions are used during anxiety episodes, and if the provided data result to actually be useful.

2.2 Key Co-design Findings

Three intervention scenarios emerged from Phase I as priorities for the system to address:

Scenario A: Improving Clinical Conversations. Participants identified that limited appointment time and reliance on patient recall constrain therapeutic discussions. Objective emotion data collected between sessions could enrich clinical conversations and support treatment planning.

Scenario B: Breaking Anxiety Loops. Acute anxiety episodes often escalate through positive feedback loops. Participants expressed desire for real-time interventions that could interrupt these spirals before they intensify.

Scenario C: Supporting Timely Help-Seeking. The median delay between anxiety symptom onset and treatment-seeking is approximately 8 years in 60% of the cases [3]. Participants saw potential for the system to provide graduated guidance toward professional support for a timely intervention.

A critical insight emerged across phases: participants consistently rejected pathology-focused framing. Rather than a tool that monitors "anxiety," participants wanted one reflecting their complete emotional lives. This led to the four-color "emotion zone" vocabulary (green, yellow, red, blue) that tracks the full emotional spectrum rather than pathologizing negative states. This anti-stigma design principle emerged as potentially central to sustained engagement.

3 System Description

MEMOPAD comprises three interconnected components: a smartwatch application for in-the-moment support, a mobile companion app for self-reporting and data visualization, and a clinician web dashboard for longitudinal review. All components share a common emotion vocabulary based on the Valence-Arousal-Dominance (VAD) model [8], mapped to four user-facing emotion zones.

Rather than displaying raw VAD coordinates [1] or clinical labels, MEMOPAD presents emotional states through four color-coded zones: Green (e.g. Calm, Relaxed, Content, Focused), Red (e.g. Panic, Terror, Enraged, Inconsolable), Yellow (e.g. Nervous, Frustrated,

Worried, Excited), Blue (e.g. Tired, Lonely, Bored, Sad). This approach emerged from co-design as more accessible and less stigmatizing than clinical terminology. This framework tracks the full emotional spectrum, including positive states, rather than focusing exclusively on anxiety or distress.

3.1 Smartwatch Application

The smartwatch app runs Android Wear OS and provides five core features designed for Scenario B (breaking anxiety loops) and defined as 'Gentle Nudges':

- **Breathing Exercise.** A guided breathing exercise with animated visual feedback. The circular progress indicator provides a focal point during acute anxiety moments.
- **Timestamp Capture.** One-tap creation of a "Moment of Interest" that syncs to the mobile app.
- **Quick Message.** Pre-configured messages to three designated contacts (e.g., carer, partner, therapist) with 'already-sent' reminders to mitigate compulsive messaging during anxiety episodes.
- **Positive Content Carousel.** Rotating display of user-uploaded images, text affirmations, and audio clips.
- **Step-by-Step Grounding Exercise.** A guided sequence providing small and achievable steps for reaching simple goals prompt as an anxiety interruption technique.

3.2 Mobile Companion Application

The mobile app serves as the primary self-reporting and data visualization interface for personal use. A key value relays on the co-existence of self-reported content and passively collected data:

- **Emotion Wheel.** A radial interface displaying 20 discrete emotions across four zones.
- **Body Map.** A body shape where users indicate physical sensations using zone colors.
- **VAD Sliders.** Three sliders for valence, arousal, and dominance.
- **Trigger Categories.** Seven predefined categories (work, relationships, health, finances, family, thoughts, environment)—with the chance to add personalised ones—for contextualizing emotional states.
- **Lifestyle Reflection.** Sleep quality/hours, exercise, medication adherence, mindfulness practice, and optional menstrual cycle tracking.
- **Daily Timeline (Journal Screen).** Vertical timeline with 15-minute notches displaying physiological signals-based emotion detections as color-coded blocks. Confidence-based threshold filters high-confidence detections from uncertain readings. In parallel, "Moment of Interest" cards expand to show associated journal entries.
- **Calendar View (Insights Screen).** Monthly overview with dominant zone indicated per day.
- **Chart Screen.** Aggregated weekly data visualizations including zone distribution, medication tracking, and trigger frequency.
- **Summary.** Work-in-progress, data collected in monthly or longer periods of time presented as a brief paragraphs generated by a RAG model. This is the most guided way to provide

Table 1: Overview of MEMoPAD Co-design Phases

Phase	Research Question	PPIE Tools	Participants
Phase 0 (Consultation)	What shapes effective co-design with anxiety populations?	Recruiting material (e.g. flyers, surveys)	8 patients, 6 clinicians
Phase I (Pre-design)	What challenges and unresolved needs do stakeholders face?	Online survey	15 patients, 3 carers, 6 clinicians
Phase II (Generative)	How can emotion recognition technology benefit the 3 scenarios?	Custom storyboard templates	12 patients, 4 carers, 4 clinicians
Phase III (Prototyping)	How should the digital tool look and behave (i.e. UX/UI)?	Custom mockup templates	15 patients, 2 carers, 2 clinicians
Phase IV (Demo feedback)	Does the app demo meet stakeholder needs?	Not-working prototype testing	16 patients, 4 carers, 5 clinicians
Phase V (Lab testing)	Does the working prototype function as intended?	Controlled lab sessions	Ongoing (16 lab participants so far)
Phase VI (Longitudinal)	How does the system perform in daily life?	In-the-wild deployment	Planned for mid-2026

insights from data. Optionally, a correlation chart is available for users who prefer to have a graphical representation of their trends over time.

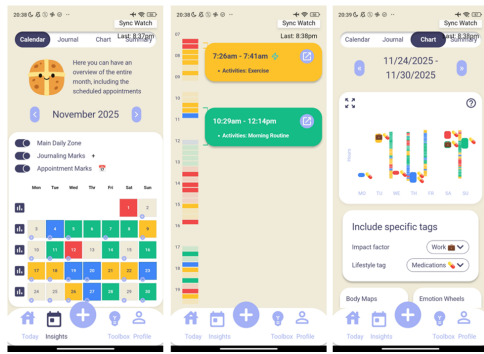


Figure 3: Screenshots of the Mobile Companion Application visualizing timeline and insights.

3.3 Clinician Web Dashboard

The clinician dashboard allows professionals to visualise the data shared by patients to support Scenario A (clinical conversations). The data visualisation options are the same available for the mobile app (i.e. personal use), as MEMoPAD is designed to be a patient-centric system. Specific features include:

- **Patient List.** Overview of consenting patients with status badges indicating recent activity levels.
- **Clinician Notes.** Secure annotation attached to specific dates or moments, supporting continuity across sessions.

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